Complex energy learning from finite magnetic-rod observations: coefficient error, local KAM and power guarantees

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Anonymous authors

Abstract

Learning an energy from simulated motion does not by itself establish the derivative accuracy needed for Hamiltonian stability or the power inequalities needed for control. We study a complex-valued network with separate configuration-energy and nonnegative dissipation heads for a finite-element magnetic rod with known constant inertia. A deterministic observation interface converts paired short trajectories at eleven configurations into intervals for 31 quadratic, cubic and quartic energy coefficients under an explicitly declared quadratic material family. A coefficient-guided objective and a precision-weighted final-readout step fit the same trained network. Outward evaluation of the saved parameters verifies all 33 finite-node force and 31 coefficient constraints, with a certified coefficient-loss upper bound below 0.010821. The conservative three-coordinate candidate satisfies a qualitative local elliptic KAM criterion. Under the declared quadratic constitutive hypothesis, a separate calculation proves this qualitative conclusion for the three-coordinate restriction of every admitted target-family member. Full-coordinate power, coercivity and dissipation are verified separately. A uniform potential-value bound admits all 22 recorded numerical initial states into a fixed energy guard. A sampled-work theorem extends conditional safety to held magnetic inputs through a strict energy ledger; nonzero initial proposals pass at both 1 and $10\,\mu$s for all 22 states. These conclusions require the stated physical family, sensing and actuation conditions. They establish neither unknown weak-nonlinearity identification, perturbative transfer from the learned Hamiltonian to an unknown plant, physical closed-loop implementation nor an advantage over comparable real-valued networks.

Keywords

complex-valued neural networks, Hamiltonian learning, magnetic rods, deterministic error bounds, local KAM theory, dissipative control

1 Introduction

An energy model learned from simulated motion can support a mathematical guarantee only when the observations, learned function and downstream dynamics are defined consistently. Magnetic actuation makes this issue relevant to deformable robots: an applied field supplies mechanical work through a configuration-dependent port, while elasticity and dissipation govern the resulting motion. Magnetic manipulation and ferromagnetic soft continuum robots provide established settings for such interactions (Abbott et al. 2020; Kim et al. 2019). The question studied here is how finite observations can constrain a learned energy sufficiently to support explicitly stated conservative and dissipative conclusions.

Hamiltonian learning supplies a useful structural starting point. A conservative vector field obtained by differentiating a scalar preserves that scalar along its exact flow (Greydanus et al. 2019). This identity does not establish that the scalar approximates a specified mechanical energy. Finite trajectory observations also leave unresolved dependence away from the observed configurations, and projection of a spatial state does not generally produce an autonomous reduced system. A learning guarantee must therefore retain observation error, the chosen mechanical chart and the distinction between a full-field model and its conservative restriction. The difficulty is especially acute when the nonlinear coefficients of interest are smaller than the uncertainty propagated from the observations.

We study a configuration potential with known kinetic energy. The mechanical object has 64 free transverse coordinates, a constant mass matrix containing translational and rotary inertia, a prescribed magnetic port, and a nominal dissipative matrix. A fixed three-dimensional trial space defines a separate conservative Hamiltonian restriction. The

learned full-field correction depends on three projected coordinates and the full nominal elastic energy; on the trial space these four features depend on only three independent configuration variables. Two complex networks represent the potential correction and a nonnegative dissipation factor, with 918 real trainable parameters in total. The kinetic energy remains fixed, so the learned correction has no momentum dependence.

The observation interface retains the full spatial records. Eleven prepared configurations each have two signed-velocity records, giving 22 selected records. Their time samples, configurations, velocities and prescribed fields produce conservative-force labels and signed weak residuals. Continuous enclosures receive each force label against the declared equations initialized by its own recorded numerical state. The bounds retain reconstruction defects, preparation error and arithmetic error. They concern a specified semidiscrete mechanical system; they do not identify a numerical reconstruction with a compiled simulator equation or turn simulated observations into physical measurements.

An exact finite design then connects the 33 force components to the 31 ordinary potential coefficients of degrees two through four. Its error bound propagates componentwise force allowances through an exact left inverse and retains an independent fifth-derivative remainder for a general target. The instantiated calculation uses an explicitly declared quadratic material-correction family, for which this remainder vanishes. This family assumption is separate from the observed labels. The coefficient objective uses the transformed label centers and their received scales, allowing its actual nonzero loss to enter a deterministic candidate-to-target coefficient bound. This finite-data route addresses the nonzero-error question raised by Chen et al. (2022) without assigning an independent sampling law to correlated trajectory data.

The saved candidate is obtained through full-layer training followed by precision-weighted calibration of its final energy readout. With the trained hidden layers fixed, the energy skip and the final complex readout form a 25-dimensional linear fitting block. One numerical least-squares solve calibrates this block within the unchanged architecture. Independent outward evaluation of the saved graph verifies all 33 force and all 31 coefficient interval constraints and supplies a strict upper bound 1.082009×10^{-2} for the nonzero coefficient loss. The same conservative restriction separately satisfies a qualitative local elliptic KAM criterion. Under the explicit hypothesis $\mathcal{W}_* = \beta E_0$, a direct interval calculation also verifies the local criterion for every target Hamiltonian in the admitted scalar interval. This second result uses the declared target family rather than a perturbative comparison with the learned candidate. For the full-field system, algebraic power identities and a uniform potential-value allowance give a conditional continuous-feedback guard. All 22 saved numerical initial states satisfy that fixed guard. A further work bound for held magnetic inputs yields a conditional sampled-feedback theorem with a strict energy ledger; all 22 nonzero initial proposals are accepted at both 1 and 10 μs hold durations.

The contribution is a connection between finite observations, an actual nonzero learning objective and conclusions for one specified saved model, with their different hypotheses kept explicit. Its limits are also informative. Exact interval intersection shows that the observations admit a scalar stiffness correction, while their quartic uncertainty exceeds the weak prescribed magnetic quartic. The data therefore do not require a nonlinear correction or establish an advantage of complex networks. The candidate and declared-family KAM results concern separate three-degree-of-freedom conservative restrictions; neither establishes an invariant trial subspace in the full rod or a KAM statement for unknown physics. The continuous and sampled safety theorems assume the stated physical family and full-state port. The sampled result also requires valid observations at the actual hold starts, bounded physical damping, and execution of the accepted total field with the prescribed timing. No physical sensor guarantee or new closed-loop rollout is supplied. Identification of unknown weak nonlinearities, perturbative learned-to-plant transfer and physical feedback implementation remain separate requirements.

2 Related work

Hamiltonian and port-Hamiltonian learning. Greydanus et al. (2019) parameterize a scalar Hamiltonian and differentiate it to construct a conservative vector field. This separates conservation of the learned scalar from unconstrained vector-field regression. Zhong et al. (2020) extend structured learning to dissipation and external inputs through a port-Hamiltonian computation graph, using observed trajectories and differentiable integration. Desai et al. (2021) likewise use port-Hamiltonian structure to learn systems with time-dependent forcing and dissipation. These works motivate retaining energy, loss and supply as distinct components of a model. We fix the native kinetic and input operators, learn a configuration correction and a scalar dissipation factor, and verify the resulting saved functions independently of their optimizer. The exact power identity is an architectural property; accuracy relative to the specified

target enters through separate observation and potential-error bounds.

Complex network representations. Trabelsi et al. (2018) develop practical components for deep complex networks, including complex convolutions, normalization and initialization, and evaluate them on vision and audio tasks. Their results motivate complex affine representations but do not supply a mechanical identification theorem for the present network. Zhang et al. (2022) establish approximation and optimization results for specified complex-reaction networks, including a representation separation for a class of radial functions. Those statements depend on their network and activation assumptions. Our split-tanh implementation is interpreted as a real differentiable graph before continuation in complex configuration variables. Its analytic domain and low-order coefficients are checked for the actual saved parameters. We use neither of these representation references as a generalization bound or a proof of superiority for this architecture. The scalar comparison admitted by the present observations makes that distinction consequential.

Nonzero learning error and conservative invariant tori. Chen et al. (2022) connect nonzero Hamiltonian learning error to KAM theory through gradient-class complexity, statistical generalization and an energy sup-norm estimate. Applying such a route requires its sampling, class and regularity hypotheses. Here the time samples, signed pairs and derived coefficients reuse a fixed finite observation design; they are not independent configuration samples. We instead propagate explicit force allowances to a finite coefficient error and then evaluate the candidate's conservative conditions separately. Classical KAM theory requires arithmetic and nondegeneracy hypotheses in addition to a perturbation estimate (Pöschel 2001). A posteriori formulations can quantify existence near an approximate torus (Haro and Luque 2019). Our instantiated result uses a local elliptic criterion with the cubic-corrected quartic normal form. It establishes qualitative candidate tori near equilibrium, without certifying a specified finite torus domain or transferring them to the target.

Magnetic robotics and the scope of mechanical verification. The survey of Abbott et al. (2020) describes magnetic forces, torques and field generation for remote robotic manipulation. Kim et al. (2019) demonstrate remotely steerable ferromagnetic soft continuum robots. These works provide the robotic context for a mechanical input port whose work must be accounted for alongside elastic storage and dissipation. The present study addresses a declared semidiscrete beam model and simulated observations. Its full-field power and conditional safety arguments retain the same mass and magnetic port, while the three-dimensional conservative restriction serves a different purpose. A verified reduced candidate, a finite-element observation record and a physical robot remain distinct objects unless their transfer errors and implementation conditions are separately bounded.

3 Mechanical object and complex energy representation

3.1 Full-coordinate finite-element mechanics

The mechanical object is the transverse finite-element model of a clamped magnetic rod. Its 32 free nodes supply $x = (w_1, \theta_1, \dots, w_{32}, \theta_{32})^T \in \mathbb{R}^{64}$, with transverse displacement and cross-section rotation at each node. The mesh has spacing 0.0025 m over a length of 0.08 m. The assembled matrices define a particular finite model; no equivalence to an arbitrary geometrically nonlinear continuum rod is assumed. The nominal elastic energy and the known inertia are

$$\begin{aligned} E_0(x) &= \tfrac{1}{2} x^T K_0 x, \\ M = M_0 + M_r \succ 0, &\qquad D_0 = \alpha M_0 \succ 0. \end{aligned} \tag{1}$$

Here M_r is the rotary-inertia contribution, $K_0 \succ 0$ and $\alpha > 0$. The damping matrix uses M_0, rather than the total mass. All quantities in this section use physical SI time and energy. The exact saved finite-element entries define the reference operators; the sum $M_0 + M_r$ is formed before any further rounding.

Let $b = e_{64}$ select the tip rotation. For a magnetic field u in tesla and fixed positive coefficients κ and μ, define

$$\begin{aligned} U_0(x) &= E_0(x) + \kappa[1 - \cos(b^T x)], \\ g(x) &= \mu \cos(b^T x) b. \end{aligned} \tag{2}$$

The learned full-state equations are

$$\dot{x} = M^{-1} p = v, \qquad \dot{p} = -\nabla U_N(x) - D_N(x) v + g(x) u. \tag{3}$$

The simulation target used for coefficient evaluation has correction $\mathcal{W}_*(x) = \beta E_0(x)$ and constant damping proportional to D_0. This explicitly declared quadratic family provides a constitutive assumption for the numerical coefficient receiver. It is not inferred to be a general unknown physical law.

3.2 Restriction and its closure boundary

Fix a full-column-rank preparation matrix $V \in \mathbb{R}^{64\times 3}$ and define

$$\begin{aligned} A &= V^T M V, \qquad B = A^{-1} V^T M, \\ K_3 &= V^T K_0 V, \qquad d = V^T b. \end{aligned} \tag{4}$$

Thus $BV = I_3$ and $q = Bx$ is the observed configuration feature. Restricting $x = Vq$ and pulling back the canonical form gives the three-degree-of-freedom Hamiltonian

$$H_N^r(q, p_r) = \tfrac{1}{2} p_r^T A^{-1} p_r + U_N(Vq). \tag{5}$$

The associated full momentum is $p = MVA^{-1}p_r$. This is a Hamiltonian restriction with a well-defined canonical structure. It is not a proof that $\mathrm{range}(V)$ is invariant under the full 64-coordinate dynamics. Full-field observations retain their full coordinates in the label construction and in the power analysis; they are not replaced by a projected autonomous trajectory.

3.3 The two complex-valued heads

Both heads are fully connected $4 \to 12 \to 12 \to 1$ complex networks. For a hidden complex value z, the split activation is $\sigma(z) = \tanh(\mathrm{Re}\, z) + i \tanh(\mathrm{Im}\, z)$. Affine weights and biases are complex; the physical output energy is real. This implementation follows complex affine components and real/imaginary processing as in Trabelsi et al. (2018). Its realification, rather than a claim that a real-valued energy is holomorphic in the original complex hidden variable, defines the analytic continuation used below.

With saved positive scales Q_j and E_{net}, the four real features are

$$s(x) = \big(q_1/Q_1, q_2/Q_2, q_3/Q_3, E_0(x)/E_{\mathrm{net}}\big), \qquad q = Bx. \tag{6}$$

Let $N(s)$ be the real energy readout and $(R(s), I(s))$ the real and imaginary dissipation readouts. Set $\gamma_j = \partial_j N(0)$ for $j = 1, 2, 3$ and define

$$F_N(x) = c_E E_0(x) + E_{\mathrm{net}} \mathcal{N}_c(s(x)), \tag{7}$$

$$\mathcal{N}_c(s) = N(s) - N(0) - \sum_{j=1}^{3} \gamma_j s_j,$$

$$\begin{aligned} a_N(x) &= [1 + c_C + R(s(x)) - R(0)]^2 \\ &\quad + [I(s(x)) - I(0)]^2, \end{aligned} \tag{8}$$

$$U_N(x) = U_0(x) + F_N(x), \qquad D_N(x) = a_N(x) D_0. \tag{9}$$

The two branches and the real scalars c_E, c_C contain 918 real parameters. Equation (7) enforces $F_N(0) = 0$ and $\nabla F_N(0) = 0$. Only the first three feature derivatives are subtracted. The linear response through the energy feature is retained and contributes to the physical quadratic term. Equation (8) enforces $D_N \succeq 0$ for every parameter vector but does not alone guarantee a positive lower bound.

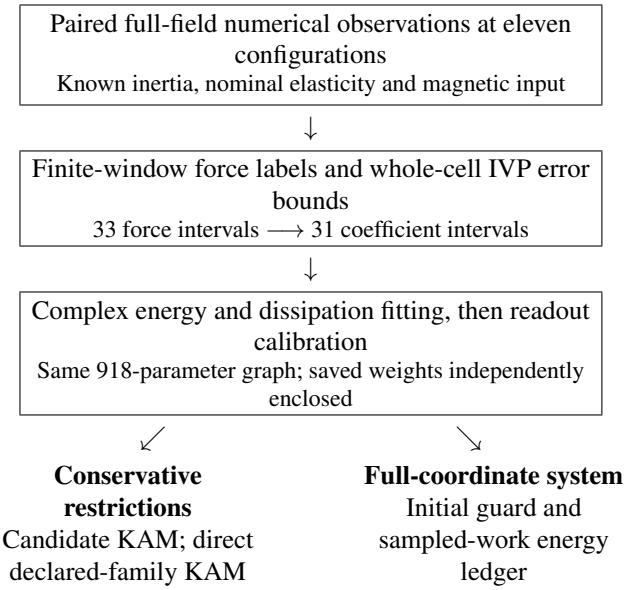


Figure 1. The observation-to-proof chain. Conservative KAM concerns the three-degree-of-freedom restrictions with forcing and dissipation removed: the learned candidate and, under $\mathcal{W}_* = \beta E_0$, every admitted target-family member. Full-coordinate safety uses a separate potential-value guard and continuous or sampled feedback under explicit physical and execution conditions. All 22 saved numerical initial states satisfy the guard; sampled calculations receive initial proposals, without executing feedback. Neither branch identifies unknown physics, certifies a physical sensor or establishes complex-network superiority.

The configuration feature and the full elastic-energy feature must both be differentiated. Writing $F(q, E)$ for the right-hand side of (7) gives

$$\begin{aligned} \nabla_x F_N &= B^T F_q + F_E K_0 x, \\ V^T \nabla_x F_N &= F_q + F_E V^T K_0 x. \end{aligned} \tag{10}$$

On $x = Vq$, $E_0(Vq) = q^T K_3 q/2$; on a general full trajectory the elastic-energy feature is not determined by Bx alone. This is why the same full observation enters the dissipative weak residual.

4 Finite observations and coefficient error bounds

4.1 Paired trajectories and finite-window force labels

The data are native COMSOL outputs at eleven prescribed configurations. Each configuration has two signed initial velocities and 257 saved full-field observations over 20 μs.

The eight added configurations use $q = \ell z^{(i)}$, $\dot q = \pm 25 q\,\mathrm{s}^{-1}$ and internal step 1.953125×10^{-8} s. The preceding three retain their own prescribed velocities and finest saved steps. These choices are documented before their respective computations. The source model is unchanged, and the drive is the source-decimal function

$$B_y(t) = 10^{-4}\cos(11.41692154656479\,t)\ \mathrm{T}. \tag{11}$$

The generator's material is public in the simulation protocol. No material factor or analytic target coefficient is supplied to the learning worker, but this separation is not a hidden-target test.

The finite momentum-and-magnetic-force label approximates a conservative force at a prepared configuration. A trajectory average cannot be identified with its initial value without a temporal allowance. For signed records, let $Y(T)$ be the average obtained from projected momentum increments and magnetic-force quadrature, with the known nominal elastic force subtracted. We use the fixed three-window combination

$$\widehat{f} = \tfrac{8}{3}Y(T) - 2Y(2T) + \tfrac{1}{3}Y(4T). \tag{12}$$

For an exact smooth signal, the constant term is preserved and the first two temporal moments cancel. If its third derivative is bounded by B_3, the remaining temporal bias is at most $B_3T^3/3$. Preparation defects and finite observation errors must also be included; they are not estimated from a small training residual. The full statement and its nonnegative Peano kernel are proved in Supplementary Section S2.1.

Continuous error is received against an explicitly declared 64-coordinate initial-value problem starting at each saved numerical first row. Its shifted elastic matrix is $K = (53/50)K_0$, and its damping is $D = \alpha_{\mathrm{shift}} M_0$, with the saved binary coefficient $\alpha_{\mathrm{shift}} \simeq 0.6389230223455499\,\mathrm{s}^{-1}$. These are the declared evaluation-IVP coefficients, not extra material labels given to the learner. A cubic Hermite curve through saved positions and velocities has a computable whole-cell equation defect. In the mechanical energy metric the trajectory-error radius obeys

$$r'(t) \le a r(t) + \rho(t), \qquad r^2 = e_x^T K e_x + e_v^T M e_v, \tag{13}$$

where ρ bounds the mass-dual equation defect. Outward Bernstein-polynomial bounds, trigonometric remainders and the exact source-decimal input enclose each whole time cell. A projected-force average retains cancellations that a generic endpoint-momentum bound would discard. The resulting received label allowances are below 4.583×10^{-8} in normalized force at all eleven chosen configurations. This certifies the declared recorded-state IVPs; it does not certify physical measurement error or equivalence to every compiled COMSOL operation. Supplementary Sections S2.3 and S2.8 give the full construction.

4.2 The eleven-node fourth-order design

Set $q = \ell z$, $\ell = 10^{-3}$ m, and use the exact decimal energy scale $E_s = 6 \times 10^{-12}$ J for the force/coefficient interface. The saved binary scale E_{net} is kept distinct from E_s. For the restricted correction $W_*(q) = \mathcal{W}_*(Vq)$, define $w_*(z) = W_*(\ell z)/E_s$. The normalized force labels approximate $\nabla_z w_*(z^{(i)})$ at the ideal rational nodes, while the learner's once-rounded input coordinates are recorded separately.

Under $\nabla w_*(0) = 0$ and a componentwise fifth-derivative bound M_5, its ordinary Taylor coefficients of degrees two through four form $c_* \in \mathbb{R}^{31}$. With the fixed gradient design matrix $G \in \mathbb{Q}^{33\times 31}$ and exact left inverse J, one has

$$\begin{aligned} JG &= I_{31}, \\ |c_{*,\alpha} - (J\widehat{y})_\alpha| &\le \sum_{i,j} |J_{\alpha,(i,j)}|\varepsilon_{ij}, \\ \varepsilon_{ij} &= \eta_{ij} + \frac{M_5}{24}\|z^{(i)}\|_1^4. \end{aligned} \tag{14}$$

Here $\widehat{y} \in \mathbb{R}^{33}$ contains the saved force midpoints and η includes each component's continuous and arithmetic allowance. The exact full-rank construction and tail derivation are in Supplementary Proposition S10. The fifth-derivative bound is an independent target hypothesis. For the explicitly declared family $W_* = \beta q^T K_3 q/2$, $M_5 = 0$ and the origin gradient vanishes by that constitutive assumption.

Writing $\widehat{c} = J\widehat{y}$, the largest received radii for ordinary coefficients of degrees two, three and four are

$$\begin{aligned} \max_{|\alpha|=2} r_\alpha &\le 4.282817 \times 10^{-7}, \\ \max_{|\alpha|=3} r_\alpha &\le 8.141379 \times 10^{-7}, \\ \max_{|\alpha|=4} r_\alpha &\le 1.808799 \times 10^{-6}, \end{aligned} \tag{15}$$

respectively, rounded upward. These are deterministic error intervals from the same 33 observations, not 31 additional independent samples. A small force error does not by itself bound arbitrary higher derivatives; the design, centering and remainder hypotheses in (14) are essential.

4.3 Nonzero-loss bounds for the saved network

Let $c_{\theta,\alpha}$ be the ordinary normalized correction coefficients of the actual saved graph. The mathematical data centers are $\widehat{c} = J\widehat{y}$. Their saved training centers $\widetilde{c}$ are binary numbers. Exported radii satisfy $\widetilde{r}_\alpha \geq r_\alpha + |\widetilde{c}_\alpha - \widehat{c}_\alpha|$, including the exact center-rounding allowance. With the fixed saved scales $\widetilde{s}_\alpha = \max(\widetilde{r}_\alpha, 10^{-8})$, the actual coefficient objective is

$$\widetilde{L}_{\mathrm{jet}}(\theta) = \frac{1}{31} \sum_{\alpha} \left(\frac{c_{\theta,\alpha} - \widetilde{c}_\alpha}{\widetilde{s}_\alpha} \right)^2 . \tag{16}$$

The data interval gives, for every parameter vector,

$$|c_{\theta,\alpha} - c_{*,\alpha}| \leq |c_{\theta,\alpha} - \widetilde{c}_\alpha| + \widetilde{r}_\alpha \leq \widetilde{s}_\alpha \sqrt{31 \widetilde{L}_{\mathrm{jet}}(\theta)} + \widetilde{r}_\alpha . \tag{17}$$

No optimizer convergence or independent-sampling assumption is used. Each component's direct residual normally gives the tighter first bound. The exact original-center version is also valid with $\widehat{c}$, r and any fixed positive scales, but is not silently substituted for the saved-center objective actually optimized.

Float64 losses guide optimization but are not proof bounds. For an outward coefficient enclosure $[a_\alpha, b_\alpha]$, define $\Delta_\alpha = \max(|a_\alpha - \widehat{c}_\alpha|, |b_\alpha - \widehat{c}_\alpha|)$. Then $|c_{\theta,\alpha} - c_{*,\alpha}| \leq \Delta_\alpha + r_\alpha$. Replacing $\widehat{c}_\alpha$ by $\widetilde{c}_\alpha$ in the endpoint residual gives $\widetilde{\Delta}_\alpha$; then $(1/31)\sum(\widetilde{\Delta}_\alpha/\widetilde{s}_\alpha)^2$ is a rigorous upper bound for the actual saved-center loss. Supplementary Section S3.3 proves this finite-loss interface and distinguishes original from rounded training centers.

The configuration-potential formulation also permits a conditional population-risk route inspired by Chen et al. (2022). That route requires independent configuration samples, a justified bounded loss class, a density lower bound and target regularity. The eleven deterministic preparations do not instantiate those statistical assumptions. The present numerical conclusions use (14)–(17); the separate population estimates and their prerequisites are retained in Supplementary Section S1.

5 Training and precision-weighted coefficient fitting

5.1 Supervision of energy and dissipation

The energy head is supervised by the eleven observed conservative force labels and their derived coefficient intervals. Its point prediction is the normalized gradient of $F_N(Vq)$, including the elastic-energy feature derivative in (10). The dissipation head uses the odd part of paired weak momentum residuals on the 22 full-coordinate records. Each signed record supplies four 64-panel windows. An individual weak residual includes momentum endpoints and an integral of conservative, dissipative and magnetic forces. The signed difference emphasizes the velocity-dependent term but retains the actual imperfect initial symmetry and numerical observations. It is not assumed to be an exact isolated damping measurement.

The conservative contribution is detached during dissipation updates. Thus energy and dissipation are fitted by specified block objectives; the whole procedure is not described as complete gradient descent on a single coupled selection score. All eleven configurations are development data. Neither their 257 temporal samples nor their velocity signs are counted as independent configuration samples.

5.2 Full-layer fitting and its high-order failure

The network is initialized from a preceding single-configuration fit that performed 200 Adam updates and selected step 191. Eleven-configuration full-batch training then performs 2000 Adam updates at learning rate 10^{-3} and seed 43, selecting step 2000 by the combined same-data score. Both real and imaginary parts of all three weight layers in both branches change. The maximum normalized point-force residual is approximately 1.563110×10^{-5}. Direct outward evaluation nevertheless finds a quartic coefficient discrepancy larger than 6.345877×10^{-4} against the declared quadratic material family. This comparison is made after training; analytic zero cubic or quartic values are not substituted as labels.

Adding (16) reduces that high-order discrepancy. The energy branch uses one strong-Wolfe LBFGS iteration per outer cycle, and the dissipation branch one Adam step at fixed current energy. All 200 prescribed outer cycles are completed, with 432 energy closure evaluations and 200 dissipation updates. The coefficient loss falls from about 50743.76 to 4.399638. However, the point-force residual rises to approximately 2.434500×10^{-4}. In this stage the point residual is divided by the observed signal RMS whereas the coefficient residual is divided by its much smaller received error. A small coefficient objective therefore does not ensure an adequate finite-node fit under that scaling.

5.3 A linear block in the unchanged complex graph

Fixing the two trained hidden layers makes the energy correction linear in c_E and the 24 real coordinates of its final complex weight. The centered output bias is a gauge

and remains fixed. Let $\xi \in \mathbb{R}^{25}$ contain these coordinates and $A_\psi \xi \in \mathbb{R}^{64}$ contain the 33 force and 31 coefficient predictions. The final calibration solves the numerical least-squares problem

$$\min_{\xi} \left\| D^{-1} \left(A_\psi \xi - \begin{pmatrix} \widehat{y} \\ \widetilde{c} \end{pmatrix} \right) \right\|_2^2, \qquad D = \operatorname{diag}(\eta, \widetilde{s}). \tag{18}$$

Here $\widetilde{c}$ and $\widetilde{s}$ are the saved coefficient centers and scales including center-rounding allowances. Each term is $(\text{residual}/\text{scale})^2$. There is no ridge penalty, and the force scales are their own received observation errors. The numerical design has rank 25 and condition estimate 2.84×10^6. These are floating-point diagnostics, not an exact-rank or exact-optimality proof. The exact-real fit is verified after saving the selected weights.

This is one final-readout solve and zero gradient updates. Only c_E and the final energy weight change; all other coordinates, including the entire dissipation head, remain identical to the preceding trained state. The model retains its 918-parameter graph. The method and scaling were chosen after observing the preceding fit and are reported as adaptive development, without a blind validation claim. The 64 rows in (18) do not represent 64 independent observations. Supplementary Section S3.4 proves the linear-block identity for the same centered network.

5.4 *Independent reception of the selected parameters*

Arb evaluation at 512 bits encloses the exact-real outputs defined by the saved binary weights, saved scales and exact native operators. At all eleven ideal rational nodes the 33 force enclosures satisfy their respective received intervals. All 31 coefficient enclosures also satisfy the original data intervals. The maximum normalized point-force residual is below 1.797797×10^{-8} and its largest ratio to its received allowance is below 0.790301. Including the admitted observation error gives the finite-node candidate-to-target force bound 5.023184×10^{-8}. This bound is restricted to the eleven configurations and to the declared observation/target assumptions.

The independently computed loss upper bound is $\overline{L}_{\mathrm{jet}} < 0.01082009$. Table 1 distinguishes data-based guarantees from the tighter evaluation available only for the declared material formula. In particular, the smaller evaluator error must not be presented as the information content of the observed coefficient intervals.

Table 1. Upward bounds on normalized ordinary correction-coefficient error. The first numerical column uses the data intervals and the triangle inequality. The second additionally evaluates the declared quadratic simulation target.

Degree	Data-based bound	Declared-family evaluation
2	4.616863×10^{-7}	3.635946×10^{-8}
3	8.174276×10^{-7}	1.768735×10^{-8}
4	2.066906×10^{-6}	3.908709×10^{-8}

The final energy skip is about 0.06002256805 and the norm of the 24 real readout coordinates is about 0.0017053353. The raw skip is not itself an identified material factor, because the neural energy-feature response also contributes to the quadratic term. An exact rational feasibility calculation admits the scalar comparison family $W(q) = \zeta q^T K_3 q/2$ with nonempty compatible interval approximately $[0.0599999981693, 0.0600000004402]$. An exact rational midpoint satisfies all 33 force and 31 coefficient constraints without assigning the material value to ζ. Thus the finite data do not require a nonlinear neural correction. Detailed source-bound coefficient and feasibility calculations appear in Supplementary Section S4.1.

6 Conservative learning error and local KAM certification

6.1 *What the coefficient bound can certify*

For the restriction (5), known kinetic and reference magnetic terms are shared between the learned and target Hamiltonians. Their vector-field difference therefore depends only on the conservative correction force. The learning variable can be three-dimensional configuration, even though the canonical phase space has six real dimensions. This does not reduce the mechanical number of degrees of freedom. Under independent configuration sampling and suitable class and regularity conditions, the configuration-risk estimates in Supplementary Section S1 provide one connection with the nonzero-loss theory of Chen et al. (2022).

The deterministic bound (17) supplies a different, finite-data interface. Multiplication by $E_s \ell^{-|\alpha|}$ gives physical ordinary monomial coefficients; the derivative $D^\alpha W(0)$ additionally includes the factor $\alpha!$. The resulting Hessian, cubic and quartic intervals could support a physical-target KAM conclusion only if they imply the required spectral, nonresonance and twist conditions uniformly, together with the target's analytic hypotheses. Without additional constitutive structure, the present quartic intervals do not establish that implication. Their radius is much larger than the largest prescribed normalized magnetic quartic

coefficient, approximately 5.363314×10^{-9}. All fifteen admitted total quartic coefficient intervals contain zero. This sign-ambiguity diagnostic is not itself a calculation of the Birkhoff twist.

6.2 A criterion for the actual saved conservative graph

The selected network can also be verified as a Hamiltonian in its own right. Suppose $U_N(Vq)$ has a realified holomorphic continuation near the origin, its Hessian is positive definite, and its positive linear frequencies obey $k \cdot \omega \neq 0$ for integer vectors $0 < |k|_1 \leq 4$. The cubic terms are removed by the homological equation. With cubic generating function χ_3, the effective quartic term is

$$H_{4,\mathrm{eff}} = H_4 + \tfrac{1}{2}\{H_3, \chi_3\}. \tag{19}$$

Its action projection has the form

$$h_N(I) = \omega \cdot I + \tfrac{1}{2} I^T T_N I, \qquad \det T_N \neq 0. \tag{20}$$

Under these hypotheses, the classical local elliptic KAM argument gives a positive-measure family of invariant Lagrangian three-tori in every sufficiently small neighborhood of the equilibrium. Supplementary Proposition S14 gives the normal-form calculation and the scaling needed to apply Pöschel (2001). The finite frequency search is not an all-integer Diophantine assertion about the equilibrium frequencies: the Diophantine frequencies of the resulting tori vary in the nonlinear frequency image.

6.3 Same-checkpoint numerical result and its scope

The selected graph is pole-free on the explicit complex polydisc $|z_j| \leq 1/64$, with $q = \ell z$. Its positive mass and Hessian, finite degree-three/four divisors and cubic-corrected twist are received afresh from the same saved parameters. The frequencies are approximately

$$(11.73745102,\ 72.26615415,\ 326.09695871)\ \mathrm{rad\,s^{-1}}. \tag{21}$$

The twist has two positive and one negative eigenvalues and determinant near -5.271108×10^{19}, excluding zero. These checks meet the preceding qualitative criterion for this same conservative restriction. They do not specify a nonzero finite torus radius or certify a trajectory as an individual torus. The radius $1/64$ is an analytic-domain bound, not a torus radius.

This result concerns the learned conservative restriction with dissipation and time-dependent forcing removed. It establishes neither perturbative transfer to a physical target, invariance of the three-coordinate subspace inside the full rod, nor torus persistence under the dissipative controller. The following direct target-family calculation uses an additional constitutive hypothesis.

6.4 A direct local result for the admitted target family

Assume explicitly $\mathcal{W}_* = \beta E_0$, with β in the entire exact scalar compatibility interval of Section 5; its endpoints are approximately 0.0599999981693 and 0.0600000004402. The conservative three-coordinate target restriction is then

$$H^r_\beta(q, p_r) = \tfrac{1}{2} p_r^T A^{-1} p_r + \tfrac{1+\beta}{2} q^T K_3 q + \kappa[1 - \cos(d^T q)]. \tag{22}$$

Its cubic part vanishes. If u_j are mass-orthonormal linear modes, the quartic twist in physical canonical actions depends on $a_j = (d^T u_j)^2/\omega_j$ and obeys

$$\det T_\beta = -\frac{5\kappa^3}{512}(a_1 a_2 a_3)^2. \tag{23}$$

Uniform interval enclosures verify a positive simple spectrum, strictly positive modal port overlaps, and all frequency divisors through order four over the complete parameter interval. They give $|k \cdot \omega| > 11.73745099\ \mathrm{rad\,s^{-1}}$ for $0 < |k|_1 \leq 4$, together with

$$-6.217103 \times 10^{17} < \det T_\beta < -6.217101 \times 10^{17}. \tag{24}$$

The twist has two positive and one negative eigenvalues throughout. Every admitted family member therefore has positive-measure families of invariant Lagrangian three-tori in sufficiently small neighborhoods of its equilibrium. Supplementary Corollary S16 supplies the uniform calculation and its analytic hypotheses.

This is a direct qualitative theorem for the declared family; it does not assert a numerical torus radius or an all-integer Diophantine property of the equilibrium frequencies. It also does not identify that family for an unknown device, prove perturbative transfer from the learned candidate, or establish invariance of the trial subspace within the full 64-coordinate system. The candidate certificate and the target-family certificate consequently answer distinct questions.

7 Full-coordinate power and conditional magnetic safety

7.1 Power identity and positive dissipation

Let $H_N(x,v) = v^T M v/2 + U_N(x)$ and let $y = g(x)^T v = \mu\cos(b^T x)b^T v$ be the full-state magnetic port. Equations (3) and (10) give

$$\dot{H}_N = -a_N(x)v^T D_0 v + uy. \tag{25}$$

This identity uses the exact constant mass and the actual full state, including the energy-feature chain rule. It holds for all network parameters and is independent of the restricted KAM result. It expresses structure preservation, not identification of the physical dissipative force. No learned momentum correction enters $\dot{x} = M^{-1}p$, so the candidate and plant ports agree if their mass, input map and full state agree.

For the selected weights, $c_E \simeq 0.06002256805$ and $c_C \simeq 0.07237264643$. The condition $1 + c_E > 0$, bounded real split-tanh output and a quadratic completion prove real coercivity of the candidate energy. Exact native factorizations give $D_0 > (6/25)M\,\mathrm{s}^{-1}$. On the full-state feature box $|s(x)|_\infty \le 2$, exact saved-weight bounds give

$$a_N(x) > 0.3977230, \qquad D_N(x) > 0.0954535M\,\mathrm{s}^{-1}. \tag{26}$$

These are same-checkpoint lower bounds. The feature box is not automatically the full configuration energy domain used below. Supplementary Section S6.1 gives the other feature radii and exact conventions.

7.2 A potential-value safety theorem for the physical plant

Consider a physical system with the same mass and magnetic map, potential U_*, locally Lipschitz damping $D_*(x) \succeq 0$, and no additional unmodeled work. Suppose on the closure of a bounded configuration domain Ω,

$$|U_N - U_*| \le \varepsilon_U. \tag{27}$$

For actual continuous feedback $u = -\operatorname{sat}_U(ky)$, $U, k > 0$, assume

$$x(0) \in \Omega, \quad H_N(0) + \varepsilon_U \le c, \quad \inf_{\partial\Omega} U_N - \varepsilon_U > c. \tag{28}$$

Then the physical trajectory exists for all positive time, remains in Ω and obeys

$$H_*(t) \le H_*(0) \le c, \qquad H_N(t) \le c + \varepsilon_U. \tag{29}$$

Indeed, $\dot{H}_* = -v^T D_* v - y\operatorname{sat}_U(ky) \le 0$. The initial value estimate gives $H_*(0) \le c$. A first exit at the configuration boundary would require $H_* \ge U_* > c$, contradicting the energy inequality. Compactness and positive mass bound the velocity and allow continuation. This proof requires a uniform potential-value allowance; a finite-node force fit alone does not supply it. Supplementary Theorem S18 states the precise smoothness and domain hypotheses and gives the complete proof.

The physical and learned storage derivatives need not coincide. Writing $G = \nabla(U_N - U_*)$ gives, along the physical plant, $\dot{H}_N = -v^T D_* v + uy + v^T G$. Consequently a learned-energy sublevel is not automatically invariant even when the physical energy decreases. The conclusion (29) uses the potential-value comparison and the physical storage, rather than assuming away this discrepancy.

7.3 A same-checkpoint conditional guard

For a numerical instance, assume the physical family

$$U_*(x) = (1+\beta)E_0(x) + \kappa[1 - \cos(b^T x)], \tag{30}$$

where the constant β lies in the exact scalar compatibility interval of Section 5. This is an explicit constitutive-family and interval assumption. Simulation-derived compatibility is not evidence that an arbitrary physical rod necessarily belongs to that family.

On $\Omega_4 = \{x : E_0(x) < 4E_{\mathrm{net}}\}$, recomputed global derivative bounds and outward origin evaluation for the selected energy head prove

$$\sup_{\overline{\Omega}_4} |U_N - U_*| < 0.003838998456E_{\mathrm{net}}, \tag{31}$$

$$\inf_{\partial\Omega_4} U_N > 4.236161085119E_{\mathrm{net}}. \tag{32}$$

Set $c = 2.05E_{\mathrm{net}}$. The boundary guard has positive normalized margin 2.182322086663. With $h_0 = 2.046161001544E_{\mathrm{net}}$, the initial set

$$\mathcal{I} = \{(x,v) : x \in \Omega_4, \quad H_N(x,v) \le h_0\} \tag{33}$$

contains the origin and an open neighborhood. Thus (29) follows conditionally for every initial state in $\mathcal{I}$ under the specified continuous full-state feedback. The global energy-head Hessian, compliance and scalar-interval calculation are supplied directly in Supplementary Section S6.2.

All 22 finest signed records at the eleven configurations belong to this same fixed initial set. Each calculation uses its own saved $t = 0$ full-coordinate position and velocity, interpreted as exact binary64 values, with no replacement by

Vq or by the opposite sign. The strict common bounds are

$$0.237377387997 < H_N(x_0, v_0)/E_{\mathrm{net}}, \quad H_N(x_0, v_0)/E_{\mathrm{net}} < 0.972247611362. \tag{34}$$

$$\frac{c - \varepsilon_U - H_N(x_0, v_0)}{E_{\mathrm{net}}} > 1.073913390182. \tag{35}$$

Supplementary Corollary S21 gives the separate sign-wise bounds and native coordinate reception. These are numerical initial states, with no certified error bound to a true physical initial state.

7.4 A sampled-work bound and strict energy ledger

For sampled control, assume additionally the physical damping form $D_*(x) = a(x)D_0$, with locally Lipschitz a and $0 \le a(x) \le 6/5$. This upper operator bound is a physical hypothesis separate from (26). Let u_k be the actual total transverse field held for duration h_k, with $|u_k| \le U$. It replaces the preparation cosine drive. On $H_* \le c$, the full-state equation and the energy cap give an output-rate bound $|\dot{y}| \le L_y$. For the present native matrices and $U = 10^{-4}$ T, outward evaluation proves

$$L_y < 49.953396682 \ \mathrm{J\,T^{-1}s^{-2}}. \tag{36}$$

At the actual hold start t_k, require a valid current observation $|y(t_k) - \widehat{y}_k| \le \varepsilon_k$. Since $\dot{H}_* = -a(x)v^T D_0 v + u_k y \le u_k y$, every hold prefix $0 \le s \le h_k$ below the energy cap satisfies

$$H_*(t_k + s) - H_*(t_k) \le F_k(s), \quad F_k(s) := s(u_k\widehat{y}_k + |u_k|\varepsilon_k) + \tfrac{1}{2}s^2|u_k|L_y. \tag{37}$$

The omitted dissipation is nonnegative. This choice takes zero dissipation credit; it does not assume a zero derivative of the physical damping coefficient.

Initialize a trusted ledger by rounding an upper enclosure of $H_N(x_0, v_0) + \varepsilon_U$ upward to a fixed energy grid g, so $H_*(t_0) \le L_0 < c$. Write $\Delta_k = F_k(h_k)$. Accept a nonzero proposal only if its observation and actual field limit are valid and

$$\Delta_k \le \alpha_{\mathrm{res}}(c - L_k), \qquad 0 < \alpha_{\mathrm{res}} < 1, \tag{38}$$

$$L_{k+1} = \lceil \max\{0, L_k + \Delta_k\} \rceil_g < c. \tag{39}$$

Otherwise apply exactly zero and retain the ledger. Starting with $H_*(t_k) \le L_k < c$, convexity of F_k yields $F_k(s) \le \max\{0, \Delta_k\}$. A hypothetical first time with $H_* = c$ would therefore satisfy $H_* \le L_k + \max\{0, \Delta_k\} < c$, a contradiction; the rate bound is used only before that first exit. At the hold endpoint, $H_*(t_{k+1}) \le L_{k+1} < c$. Induction gives the strict energy cap for any sequence of holds without finite-time accumulation, provided the observation, timing and total-field execution conditions hold at every step. Also $c/(1 + \beta_-) < 4E_{\mathrm{net}}$, so this physical cap keeps the configuration in Ω_4. Supplementary Section S7 gives the rate majorant, continuation argument and complete execution assumptions.

For a received initial-proposal calculation, fix $\varepsilon_y = 10^{-8}\,\mathrm{J\,T^{-1}s^{-1}}$, $g = 10^{-9}E_{\mathrm{net}}$ and $\alpha_{\mathrm{res}} = 1/2$. At each of the 22 saved numerical states, an outward port enclosure supplies a rational midpoint $\widehat{y}_0$ whose arithmetic error is below this allowance. The nonzero opposite-sign proposal $u_0 = -U\,\mathrm{sign}(\widehat{y}_0)$ passes (38)–(39) for all 22 states at both $1\,\mu$s and $10\,\mu$s. At $100\,\mu$s, all 22 proposals are rejected and the exact zero fallback retains their initial ledgers. The same cap, field limit, error allowance and rounding grid are used throughout; Supplementary Section S7.1 gives the corresponding ledger and work bounds.

The port-error allowance is conditional for a physical sensor, and the calculation executes no feedback rollout. Accepted holds may spend energy reserve; neither monotone energy decrease nor improved control performance is asserted. The arbitrary-horizon theorem depends on valid sensing at each actual hold start and execution of each accepted total field with the specified timing. Physical sensing, actuator and timing guarantees remain to be supplied for an experiment.

8 Discussion

The received intervals give a precise interpretation to the fit of the selected complex model. At eleven ideal design configurations, the saved potential satisfies all 33 componentwise force constraints and all 31 coefficient constraints derived from them. Independent outward evaluation also bounds the actual coefficient loss without replacing it by zero. These results connect finite observations to one specified function. They do not require optimizer convergence, exact least-squares optimality or independent trajectory rows.

The observation precision limits what can be inferred about nonlinear constitutive behavior. The initial eleven-configuration point-force fit has a certified quartic discrepancy exceeding 6.3458×10^{-4} despite its small training score. Coefficient-guided training and final readout

calibration substantially reduce that discrepancy, but the data-based quartic error allowance remains approximately 2.067×10^{-6}. The largest prescribed magnetic quartic is only approximately 5.364×10^{-9}, and every received total quartic interval contains zero. This sign ambiguity is an information diagnostic, not a calculation of the Birkhoff twist. Exact intersection of all force and coefficient inequalities also admits a scalar stiffness correction. Thus the finite data cannot require a nonlinear correction, even when a nonlinear network satisfies every received interval.

The calibrated readout should be interpreted in light of that ambiguity. The final step changes the energy skip and 24 real coordinates of the last complex energy weight while preserving the trained hidden layers and dissipation branch. It performs one linear block solve and no gradient update. The readout norm decreases markedly, and the resulting function is compatible with a nearly scalar correction. This outcome preserves the 918-parameter function graph and its preceding full-layer training, but it provides no evidence that the complex representation is needed by these observations. The 31 coefficient rows are deterministic transforms of the same 33 force components, so their joint use in a weighted objective does not create 64 independent observations. All stages use development data; there is no independent validation or statistical generalization claim.

The deterministic loss bound also separates observation information from constitutive assumptions. The coefficient centers are obtained from the observed force labels, with their propagated errors and exported-center rounding retained. Vanishing cubic or quartic corrections are not added as training labels. The zero fifth-derivative remainder in the instantiated target bound follows from the explicitly stipulated quadratic material family. For an unknown constitutive law, that remainder and any true-state measurement errors would have to be bounded independently. A finite coefficient bound alone supplies neither a uniform analytic error norm nor the target's nondegeneracy conditions.

The conservative certificate has a correspondingly specific scope. The same saved candidate has a positive equilibrium Hessian, the required finite-order frequency gaps and a nonsingular cubic-corrected quartic Birkhoff twist. Together with its verified analyticity, these conditions give positive-measure families of invariant three-tori in sufficiently small neighborhoods of its equilibrium. They give no numerical radius for such a neighborhood. A separate calculation, conditional on $\mathcal{W}_* = \beta E_0$, verifies the local KAM conditions uniformly for every member of the admitted scalar interval (Supplementary Corollary S16). It uses the target family's analytic twist rather than the learned twist. This direct family theorem does not establish a perturbative transfer from the candidate or identify unknown physical constitutive behavior. Neither result includes damping or time-dependent magnetic forcing, or proves that the three-coordinate trial subspace is invariant in the full rod.

The full-field safety statement uses different information. The candidate has an exact port-power identity, nonnegative dissipation and a separately checked sufficient coercivity condition. For a physical plant sharing the stated mass and full-state magnetic port, a uniform potential-value allowance transfers an energy guard under actual continuous saturated feedback. The instantiated quadratic-family guard uses $E_0 < 4E_{\mathrm{net}}$, cap $2.05E_{\mathrm{net}}$ and potential allowance $0.003838998456E_{\mathrm{net}}$. All 22 recorded numerical initial states satisfy this unchanged guard, with $0.237377387997 < H_N/E_{\mathrm{net}} < 0.972247611362$ and initial margin greater than $1.073913390182E_{\mathrm{net}}$ (Supplementary Corollary S21). This numerical reception supplies no error bound to a true physical initial state. Family membership is a conditional physical assumption, not a parameter identification conclusion. Likewise, the candidate's dissipation lower bound on its feature box is not a bound on unknown physical damping or on a different configuration domain.

The sampled-work theorem supplies a sufficient condition for held inputs. With the additional physical bound $D_* = a(x)D_0$, $0 \le a \le 6/5$, it bounds intersample port drift and charges held-input work to a strict energy ledger. At the fixed 10^{-4} T total-field limit, all 22 initial nonzero proposals pass for both 1 and 10 μs holds; none passes for 100 μs. The latter rejection is a decision of this sufficient guard, not evidence of unsafe motion at that duration. No feedback trajectory is executed. The arbitrary-horizon energy conclusion requires the stated sensing, timing and actual-command conditions at every subsequent hold; a finite set of accepted initial proposals does not verify those conditions. In particular, the $10^{-8}\,\mathrm{J\,T^{-1}s^{-1}}$ port allowance has been checked for numerical arithmetic, not for a physical sensor.

The distinction between full-field observations and the conservative restriction remains essential throughout. A projected spatial trajectory can retain unresolved elastic and dissipative forces, so the three-dimensional candidate is not asserted to close every full-field motion autonomously. The force-label reception controls the particular observation interface used here. It does not establish a general trajectory approximation theorem, continuum discretization accuracy or equivalence between a stored reconstruction and all operations of a compiled simulator. The certificates interpret

the saved binary weights and mechanical coefficients as exact constants in their specified graphs; rounded runtime evaluation is a separate numerical object.

Extending these results to an unknown device therefore requires both more informative observations and an implementation contract. A useful configuration and velocity design should resolve the coefficient signal after propagating observation errors, while independent records and constitutive remainder bounds should test behavior beyond the fitting design. Physical use of the safety theorem additionally requires sound state and port estimates and control of unmodelled work. The present sampled theorem supplies an intersample work and ledger analysis under explicit conditions; physical use must establish those conditions, including observation errors at the actual hold starts and execution of the accepted total field. Delays or actuation errors outside this contract need additional bounds. These are concrete conditions for transferring the present model-level results to a robotic experiment.

Data and code availability

The accompanying reproducibility manifest identifies the native simulation records, selected network parameters, exact operator records and interval verification sources used for each numerical statement. It separates local source files from workspace evidence and records their SHA-256 identities. The supplementary material supplies the mathematical proofs and arithmetic conventions. Simulation data are not represented as physical measurements, and no independent test dataset is claimed.

Declarations

This study reports mathematical analysis and simulated mechanical data. It involved no human participants or animals.

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